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A multi-dimensional instance weighting and dynamically supervised signal selection method for low-resource cross-lingual summarization

Yongbing Zhang [a,b], Fang Nan [a,b], Shengxiang Gao [a,b], Yuxin Huang [a,b], Kaiwen Tan [a,b,*], Zhengtao Yu [a,b,*]

[a] Faculty of Information Engineering and Automation, Kunming University of Science and Technology, Kunming, Yunnan, 650500, China
[b] Yunnan Key Laboratory of Artificial Intelligence, Kunming University of Science and Technology, Kunming, Yunnan, 650500, China

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ABSTRACT

Cross-lingual summarization (CLS) is an advanced natural language processing task that involves generating a summary in a language different from the source text. Recent studies have utilized the round-trip translation strategy on source language summaries from monolingual summarization datasets to construct CLS datasets, achieving success in rich-resource languages. However, for low-resource languages, the quality of the generated target language reference summaries is often compromised due to subpar machine translation performance. Applying stricter filtering thresholds can result in higher data quality but a smaller dataset size, while using lower thresholds may lead to a larger dataset but with reduced data quality. An exploratory experiment suggested that a large amount of low-quality data might be more beneficial than a small amount of high-quality data. Nevertheless, there is currently limited research on how to effectively utilize a large amount of low-quality data for training CLS models. To address this, we propose a Multi-dimensional instance weighting and Confidence-based token-level mixup method for low-resource Cross-Lingual Summarization (MCCLS), which allows for more refined utilization of large-scale and low-quality data. Specifically, a multi-dimensional instance weighting method is developed to evaluate the quality of an instance from multiple perspectives, which assigns higher weights to high-quality instances while retaining low-quality instances. Additionally, a dynamically supervised signal selection method dynamically determines supervisory signals at the token-level, thereby effectively leveraging valuable parts of low-quality instances. Experiments demonstrate that MCCLS outperforms existing models on large-scale and low-quality dataset, achieving higher ROUGE scores and stable performance across different quality ratios. The generated summaries are also more informative, fluent, and concise.

1. Introduction

Cross-lingual summarization (CLS) is intended to extract key information from a source language text and organize it into a summary in the target language (Wang et al., 2022b; Zheng et al., 2023). Due to the involvement of different input and output languages, automatically constructing large-scale CLS datasets is more challenging than monolingual summarization. To address this issue, Zhu et al. (2019) proposed using a round-trip translation strategy to generate CLS datasets from existing monolingual summarization datasets. This process involves obtaining target language summaries through forward translation and then assessing their quality through back-translation. For rich-resource languages such as Chinese and English, which benefit from abundant monolingual corpora and robust machine translation performance, this strategy can generate large-scale and high-quality CLS datasets. However, for low-resource languages like Vietnamese and Burmese, this strategy is less effective. The primary reason is the relatively poor performance of machine translation for these languages (Singh & Singh, 2022; Wang et al., 2025), resulting in low-quality target language summaries during the forward translation process. Furthermore, significant discrepancies between the back-translated content and the original summaries lead to a large number of instances being filtered out, ultimately resulting in a limited number of high-quality CLS instances. Although adjusting the filtering threshold can help retain more instances, it also introduces a substantial amount of low-quality instances into the dataset.

From the above analysis, a natural question arises: **is it more beneficial for model training to have a small amount of high-quality CLS instances or a large amount of low-quality CLS instances?** To explore this question, we designed a simple experiment: we trained the same model using three datasets of varying quality and quantity,

* Corresponding authors.
E-mail addresses: 20250026@kust.edu.cn (Y. Zhang), nanfang@stu.kust.edu.cn (F. Nan), 20170003@kust.edu.cn (S. Gao), 20220027@kust.edu.cn (Y. Huang), kwtan@kust.edu.cn (K. Tan), 12309208@kust.edu.cn (Z. Yu).

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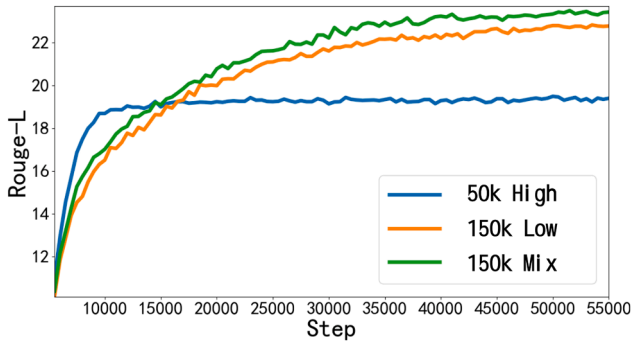


Fig. 1. Comparison of CLS model performance under different quality and quantity settings.

evaluating the model's performance under these different conditions. Specifically, we assessed the quality of each instance in the En2ZhSum training data. An instance was considered high-quality if both the ROUGE-1 and ROUGE-2 scores between its original summary and the back-translated content exceeded 0.7 and 0.35, respectively; otherwise, it was deemed low-quality. It is worth noting that when the original En2ZhSum dataset was constructed, the ROUGE-1 and ROUGE-2 thresholds were set to 0.6 and 0.2, respectively. Based on this evaluation method, we sampled 50,000 instances from the high-quality instances to create the 50k High dataset. From the low-quality instances, we sampled 150,000 instances to form the 150k Low dataset. Finally, we sampled 150,000 instances at a 1:1 ratio of high-quality to low-quality to create the 150k Mix dataset. We then trained the TNCLS model, as proposed in Zhu et al. (2019), using these three datasets.

As illustrated in Fig. 1, the Rouge-L score of the model trained on the 50k High dataset is significantly lower than that of the models trained on the 150k Low and 150k Mix datasets. Additionally, the model trained on the 150k Mix dataset slightly outperforms the one trained on the 150k Low dataset. These results suggest that, compared to small-scale high-quality datasets, large-scale datasets with low-quality instances are more advantageous for training CLS models. At the same time, the quality of the data is crucial in determining model performance when the data scale is constant. Therefore, we believe that **more refined utilization of low-quality instances within large-scale datasets is key to further enhancing the performance of CLS models, especially for low-resource CLS.**

Based on the above ideas, we propose a multi-dimensional instance weighting and dynamically supervised signal selection method for low-resource CLS (MCCLS). Specifically, we develop a multi-dimensional instance weighting method that evaluates instance quality through multiple perspectives including translation quality, factual consistency, and distribution consistency, assigning higher weights to high-quality instances while preserving the latent value of low-quality ones. Subsequently, we design a dynamically supervised signal selection method to dynamically determine token-level supervisory signals, thereby effectively leveraging valuable parts within low-quality instances. Experimental results demonstrate that MCCLS significantly outperforms existing approaches on three datasets (En2ZhSum, Zh2EnSum, and Vi2ZhSum), achieving higher ROUGE scores while maintaining stable performance across varying quality ratios. The generated summaries exhibit enhanced informativeness, improved fluency, and superior conciseness. The main contributions of this paper are summarized as follows:

- We propose MCCLS, a novel dual-level optimization framework that innovatively integrates instance-level weighting with token-level supervised signal selection, enabling effective utilization of large-scale low-quality CLS datasets for low-resource languages.
- We develop a multi-dimensional instance weighting method that systematically evaluates data quality through multiple perspective at instance-level.
- We develop a dynamically supervised signal selection method that dynamically adjusts supervision signals through real-time confidence estimation at token-level.
- We construct comprehensive experiments on En2ZhSum, Zh2EnSum, and Vi2ZhSum datasets to validate our method's effectiveness.

2. Related work

Cross-Lingual Summarization (CLS) refers to the process of summarizing a source language document and generating a summary in the target language. From data perspective, CLS suffers from a scarcity of parallel data compared to monolingual summarization, making large-scale data collection and manual annotation difficult (Cao et al., 2020). From modeling perspective, CLS requires handling both translation and summarization simultaneously (Zhu et al., 2020), making it challenging to develop models that generate accurate summaries directly.

To address these challenges, early CLS research adopted a pipeline approach, splitting the task into two subtasks: "summarize first, then translate" (Orăsan & Chiorean, 2008; Wan et al., 2010) and "translate first, then summarize" (Leuski et al., 2003; Ouyang et al., 2019). However, these methods suffer from issues such as information loss, error accumulation, and inference delays. To overcome these limitations, Zhu et al. (2019) introduced the first sequence-to-sequence model for CLS and constructed a large-scale Chinese-English and English-Chinese CLS dataset using a back-translation strategy, significantly improving performance. Similarly, Bai et al. (2021) built the English-German CLS dataset En2DeSum and proposed MCLAS, a multi-task framework with a unified decoder that generates both monolingual and cross-lingual summaries, treating monolingual summarization as a prerequisite for CLS.

In addition to pipeline (Duan et al., 2019; Zhang et al., 2016) and sequence-to-sequence models (Zhu et al., 2019, 2020), researchers have explored alternative strategies. Zhu et al. (2020) used fast alignment tools to obtain word translation distributions, allowing the decoder to generate summaries based on both output and translation distributions. Cao et al. (2020) employed dual encoders and dual decoders to jointly learn cross-lingual alignment and summarization. Nguyen and Luu (2022) demonstrated that knowledge distillation frameworks can enhance CLS performance in resource-rich settings by leveraging both labeled data and teacher models' outputs and hidden states.

With the advent of multilingual pre-trained models such as mBART (Liu et al., 2020) and mT5 (Xue et al., 2021a), researchers have explored pre-training strategies to enhance CLS. Xu et al. (2020) pre-trained CLS models using Masked Language Modeling (MLM), Denoising Auto-Encoding (DAE), monolingual summarization (MS), and machine translation (MT). Similarly, Dou et al. (2020) utilized MT and MS for pre-training, while Wang et al. (2022a) extended the CLS model based on mBART-50 by incorporating dialogue summarization tasks. These strategies have significantly improved CLS performance.

However, most CLS research has focused on resource-rich language pairs such as Chinese-English and English-German (Bai et al., 2021), relying on external resources and pre-trained models to boost performance. Research on CLS in low-resource and low-quality environments remains relatively limited.

3. Background

The MCCLS model is implemented using the Transformer encoder-decoder framework. Consider a CLS instance $\{x_i, y_i\} \in D$, where x_i denotes a source language document and y_i represents a target language summary. Initially, the encoder transforms the source language document $x_i = (x_{i,1}, x_{i,2}, \ldots, x_{i,n})$ into a hidden state representation $z_i = (z_{i,1}, z_{i,2}, \ldots, z_{i,n})$. Subsequently, the decoder generates the target language summary $y_i = (y_{i,1}, y_{i,2}, \ldots, y_{i,m})$. The encoder and decoder are

jointly trained to maximize the probability of producing the target language summary given the source language document:

$$L_\theta = \sum_{t=1}^{m} log P(y_{i,t}|y_{i,<t}, X, \theta)$$

The Transformer model architecture consists of stacked encoder and decoder layers (Vaswani et al., 2017). Each encoder layer comprises a self-attention block and a feedforward block, while each decoder layer integrates an additional encoder-decoder attention block. Both the self-attention and encoder-decoder attention blocks utilize multi-head attention to effectively capture information from latent space states across various positions. Each attention head employs a scaled dot-product attention mechanism, defined as:

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V$$

where d_k denotes the dimension of the key. In the context of multi-head attention, each head independently computes scaled dot-product attention. The input to each head consists of the original query, key, and value vectors, each transformed linearly by weight matrices W_i^Q, W_i^K, and W_i^V for the ith head. The output of each head results from the scaled dot-product attention computation are concatenated along the last dimension to form a comprehensive representation, which is subsequently projected through a linear layer to yield the final multi-head attention output. Thus, the complete computation for multi-head attention is expressed as:

$$MultiHead(Q, K, V) = Concat(head_1, head_2, \cdots, head_h)$$
$$head_i = Attention(QW_i^Q, KW_i^K, VW_i^V)$$

where W_i^Q, W_i^K, and W_i^V are learnable matrices, and h denotes the number of attention heads.

4. Methodology

To make full use of the low-quality instances within large-scale datasets, MCCLS consists of a multi-dimensional instance weighting method and a dynamically supervised signal selection method, aiming to fully exploit the value of low-quality instances from multiple different levels. The overall framework of MCCLS is shown in Fig. 2.

4.1. Multi-dimensional instance weighting

To more comprehensively evaluate the quality of an instance, the multi-dimensional instance weighting method obtains the first weight by assessing the translation quality of the target language summary using round-trip translation (Zhu et al., 2019). Next, it obtains the second weight by evaluating the factual consistency of the target language summary through textual entailment. Then, it derives the third weight by evaluating the distribution consistency between the predicted results and the actual results through Jensen-Shannon divergence. Finally, through a progressive weight fusion strategy, these three weights are incorporated into the training process of the CLS model. It is worth noting that the larger the weight, the higher the quality of the corresponding instance. Consequently, its loss will be greater, allowing for more learning from higher-quality data.

Translation quality. Round-trip translation refers to the process of translating a text from one language into another (forward translation), and then translating the result back into the original language (backward translation). This process can be used to evaluate the translation quality of an instance. Taking an example of the English-to-Chinese CLS task, given the source-target language text-summary pair (x_{en}, y_{zh}) and the source language summary y_{en} (Since existing CLS datasets are constructed from monolingual summarization corpora, they inherently contain source-language summaries.). We back-translate the target language summary y_{zh} into the source language y'_{en}. Then, using the evaluation strategy proposed by Zhu et al. (2019), we calculate the ROUGE-1 score between the source language summary y_{en} and the back-translated source language summary y'_{en}, and transform this score to obtain the weight as RTT_i for ith instance.

Factual consistency. Referring to the method proposed by Feng et al. (2022), we use textual entailment to determine the factual consistency between the target language summary and the source language summary. Specifically, we first translate the target language summary y_{zh} into the source language y'_{en}, then calculate the textual entailment score between y'_{en} and y_{en} using the RoBERTa textual entailment model. Finally, we convert the obtained textual entailment score to derive the weight TEN_i for ith instance.

Distribution consistency. We utilize Jensen-Shannon divergence (JSD) to quantify the difference between the predicted probability distribution of summary generation $p'_i = (p'_{i,1}, p'_{i,2}, \ldots, p'_{i,m})$ and the true reference summary label distribution $p_i = (p_{i,1}, p_{i,2}, \ldots, p_{i,m})$ to obtain the weight JSD_i for ith instance:

$$JSD_i = 1 - d_i$$
$$d_i = D_{JS}(p'_i||p_i) = \frac{1}{2}D_{KL}(p'_i||\frac{p'_i + p_i}{2}) + \frac{1}{2}D_{KL}(p_i||\frac{p'_i + p_i}{2})$$

where $D_{KL}(\cdot||\cdot)$ is the Kullback-Leibler divergence. $D_{JS}(\cdot||\cdot)$ is the Jensen-Shannon divergence, which can measure the distance between two probability distributions, and the value range is $[0, 1]$.

Progressive Weight Fusion. We can observe that the first two weights are independent of the CLS model, while the third weight depends on the performance of the CLS model. Therefore, in the early stages of model training, we primarily use RTT_i and TEN_i to weight the instances. As training progresses and the model's performance in generating cross-lingual summaries improves, we gradually incorporate more weighting based on JSD_i. Specifically, for the tth training step, the weight calculation process for instance i is as follows:

$$w_i = (1 - \lambda(t)) \cdot (RTT_i + TEN_i)/2 + \lambda(t) \cdot JSD_i$$

where $\lambda(t)$ is used to achieve progressive weight fusion, and its value is related to the training step t. We referred to the method proposed by Platanios et al. (2019) to calculate $\lambda(t)$ as:

$$\lambda(t) = min(1, \sqrt{t \cdot \frac{1 - I_0^2}{T} + I_0^2})$$

where I_0 represents the initial value, and T is a threshold. To more intuitively illustrate the progressive weight fusion strategy, we set $RTT_i = 0.8$ and $TEN_i = 0.6$, then plotted the variation curves of related values as the training step increases. As shown in Fig. 3, in the early stage, when the CLS model exhibits limited cross-lingual generation capability, instance weight w_i is primarily determined by round-trip translation quality (RTT_i) and factual consistency (TEN_i), which are independent of the model's predictions. This allows the model to rely on highly reliable supervision signs. In the later stage, as the model becomes more competent, the distribution consistency score (JSD_i), which reflects the quality of the model's own predictions, progressively gains influence, thereby encouraging the model to learn more from highly consistent instances.

Finally, the objective function of the CLS model with the multi-dimensional instance weighting can be rewritten as:

$$Loss_m = \sum_{(x_i, y_i) \in D} w_i log P(y_i|x_i)$$

4.2. Dynamically supervised signal selection

The core design philosophy of this method is: as training progresses, the CLS model's capabilities gradually improve, making the token se-

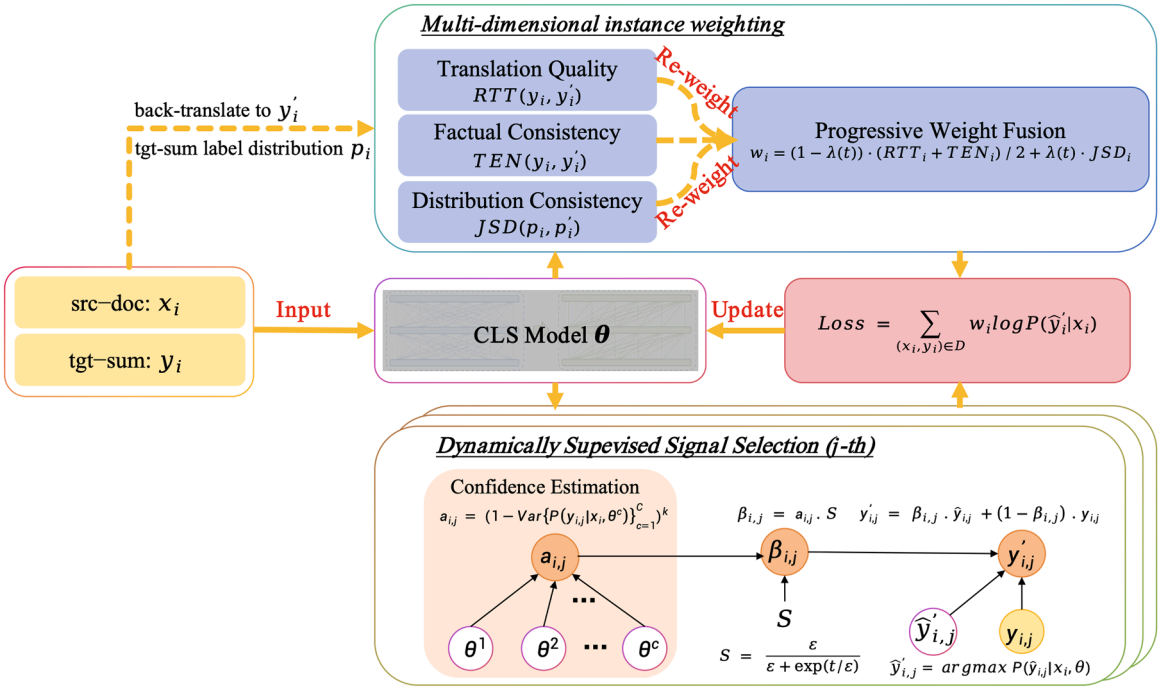


Fig. 2. The overall framework of the proposed MCCLS method.

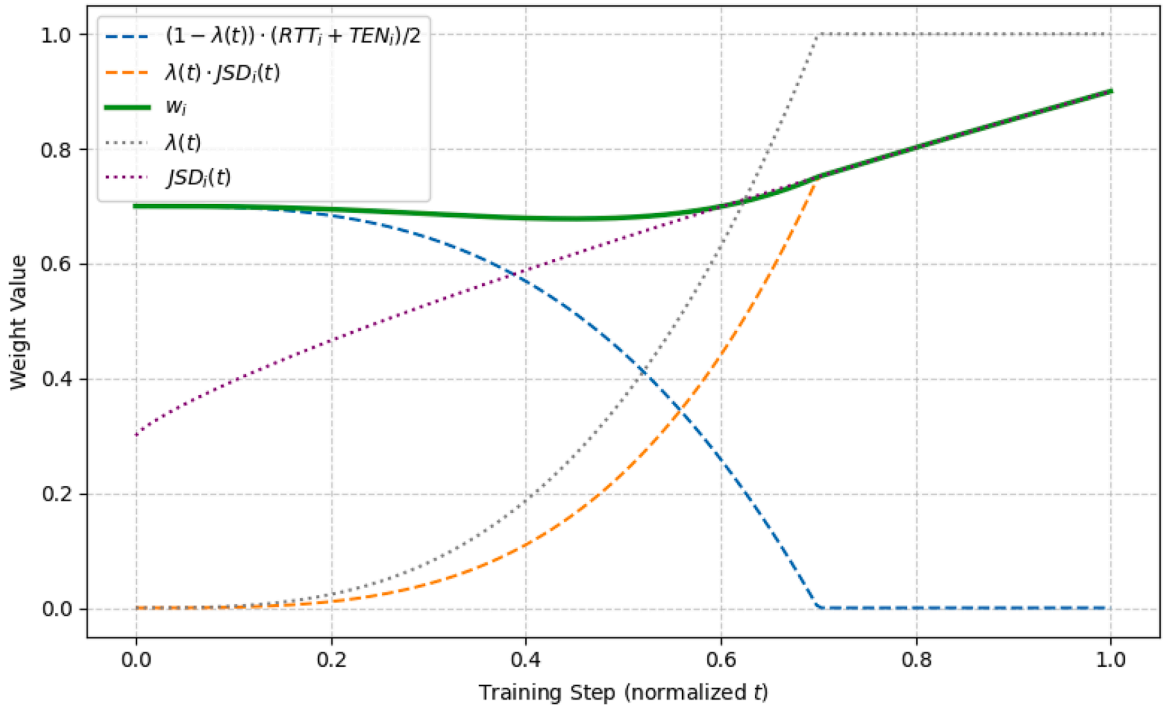


Fig. 3. Illustration of the dynamic evolution of progressive fusion weight w_i over training steps.

quences predicted by the model potentially more "reliable" (Wan et al., 2020; Zhang et al., 2019) than those in the target-language reference summaries, particularly for low-quality instances. Building on this insight, we propose a dynamically supervised signal selection approach. This method dynamically selects supervision labels for model training by evaluating the model's confidence in current output token, thereby enhancing overall training effectiveness.

4.2.1. Confidence estimation

Generally, token confidence can be quantified using Bayesian neural networks (Kwon et al., 2020). However, to improve training efficiency, we approximate Bayesian predictions through Monte Carlo sampling. Specifically, given the current parameterized CLS model and a text-summary pair $\{x_i, y_i\}$, we estimate the confidence of each token in the target-language reference summary y_i. The source-language text x_i is fed into the model C times with random neuron dropout, generating C conditional probabilities for current output token. We calculate the distribution variance from these probabilities - lower conflict probability indicates higher confidence in the training instance. The confidence estimation for jth token in y_i is formulated as:

$$\alpha_{i,j} = (1 - Var\{P(y_{i,j}|x_i, \theta^c)\}_{c=1}^{C})^k$$

where hyperparameter k controls the differentiation between confident and non-confident scores - larger absolute values indicate greater differentiation. Then, we apply softmax normalization to stabilize training and maintain conventional loss ranges:

$$\alpha_{i,j} = \frac{\exp(\alpha_{i,j})}{\sum_{t=1}^{m} \exp(\alpha_{i,t})}$$

4.2.2. Dynamically supervised signal selection

During CLS model training, our method will dynamically select reference $y_{i,j}$ or generated $\hat{y}_{i,j}$ as supervision signals based on token confidence. However, as the number of training epochs increases, the $\alpha_{i,j}$ values will grow larger, causing the model to increasingly favor the generated tokens as supervisory signals, which in turn makes it challenging for the model to continue effective training. Therefore, our reference summary selection considers both token confidence and training phase progression. Initializing with $S = 1$ indicates complete confidence dependence, with prediction-based selection frequency increasing as model confidence grows during convergence.

$$\hat{y}'_{i,j} = \beta_{i,j} \cdot \hat{y}_{i,j} + (1 - \beta_{i,j}) \cdot y_{i,j}$$
$$\beta_{i,j} = \alpha_{i,j} \cdot S$$
$$S = \frac{\varepsilon}{\varepsilon + \exp(t/\varepsilon)}$$

where ε denotes a hyperparameter, t represents training steps, and S serves as a strictly monotonic decreasing function. The ground truth weight gradually decreases throughout model training. The final objective function can be rewritten as:

$$Loss = \sum_{(x_i, y_i) \in D} w_i log P(\hat{y}'_i | x_i)$$

5. Experiments

5.1. Datasets

To verify the effectiveness of the proposed method, we conducted experiments on the CLS datasets En2ZhSum and Zh2EnSum, publicly released by Zhu et al. (2019), as well as our newly constructed Vi2ZhSum dataset. En2ZhSum is a cross-lingual English-Chinese summarization dataset containing 364,687 training instances, 3000 validation instances, and 3000 test instances. In the training set, the average length of English articles is 755 words, with an average Chinese summary length of 96 characters and an English summary averaging 55 words. Zh2EnSum is a short-text Chinese-English summarization dataset with 1,693,713 training instances, 3000 validation instances, and 3000 test instances. In the training set, the average Chinese short text is 104 characters long, with an average Chinese summary of 18 characters and an English summary of 14 words.Additionally, to evaluate the proposed method's effectiveness in low-resource and low-quality CLS settings, we constructed a Vietnamese-Chinese dataset (Vi2ZhSum) using a back-translation strategy by Zhu et al. (2019) and Google Translate. Specifically, we first collect a large number of Vietnamese article–summary pairs from news websites such as the Vietnam News Agency and Vietnam Daily Express. Then, we use the Google Translation API to forward-translate the original Vietnamese summaries S_{vi} into Chinese summaries S_{zh}. Subsequently, we apply the same translation API to back-translate the obtained Chinese summaries S_{zh} into Vietnamese summaries S'_{vi}. Following the strategy proposed by Zhu et al. (2019), we compute the ROUGE-1 score between the original Vietnamese summaries S_{vi} and the back-translated Vietnamese summaries S'_{vi}. If the ROUGE-1 score exceeds a predefined threshold $T = 0.45$, the corresponding Vietnamese article–Chinese summary pair (D_{vi}, S_{zh}) is retained; otherwise, it is discarded. Through this process, we ultimately construct the Vietnamese-Chinese CLS dataset Vi2ZhSum, which includes 129,393 training instances, 2000 validation instances, and 2000 test instances.

Table 1
Dataset sizes under different resource scenarios. Minimum, Medium and Maximum represent low-resource scenarios, medium-low-resource scenarios, and maximum-scale scenarios, respectively.

Scenarios	En2ZhSum	Zh2EnSum	Vi2ZhSum
Minimum	50,000	50,000	30,000
Medium	150,000	150,000	50,000
Maximum	364,687	1,693,713	129,393

To evaluate the model's performance under different low-resource scenarios, we randomly sampled three training sets, Minimum, Medium, and Maximum, from the complete training instances. Detailed statistics can be found in Table 1.

5.2. Experimental setup

We initialize all parameters using the Xavier initialization method and train the model based on a Transformer architecture comprising 6 encoder layers and 6 decoder layers, with a hidden dimension of 512. During training, each batch consists of a set of document-summary pairs, containing approximately 2048 source tokens and 2048 target tokens. We use the Adam optimizer (Kingma, 2014) with hyperparameters set to $\beta_1 = 0.9$ and $\beta_2 = 0.998$, and a learning rate of $\mathrm{lr} = 10^{-9}$. For confidence estimation, the hyperparameter $k = 2$ and $C = 5$. For dynamically supervised signal selection, the hyperparameter $\varepsilon = 6000$. Model training was conducted on a single NVIDIA 3090 GPU. During testing, summaries were generated using Beam Search with a beam size of 4.

5.3. Baselines

We conducted a comparative analysis of the MCCLS model with the following methods:

GETran. A "summarize first, then translate" pipeline based on Google Translate. This method first uses Google Translate to translate the source language text into the target language, followed by the widely used unsupervised summarization method, LexRank (Erkan & Radev, 2004), to generate the summary.

GLTran. A "translate first, then summarize" pipeline. This method first trains a summarization model using a monolingual summarization dataset in the source language, generates a summary in the source language, and then uses Google Translate to translate this summary into the target language. Summarization models are trained in English, Chinese, and Vietnamese, respectively, based on monolingual datasets.

TNCLS. The first end-to-end cross-lingual summarization model proposed by Zhu et al. (2019). To manage model complexity, this study uses random initialization and evaluates the model on different cross-lingual summarization datasets.

CLS + MS. A multi-task learning model based on TNCLS. It incorporates a monolingual summarization model into the traditional end-to-end framework for joint training.

In recent years, several models have improved CLS task performance. However, to ensure a fair and focused evaluation under the low-resource and noisy-data conditions central to our study, we have carefully selected baselines that align with these constraints. Specifically, we exclude models that rely on external resources or introduce additional complexity, as they would compromise the goal of assessing inherent model capabilities without external advantages. For instance: The NCLS + MT and CLS + MS methods (Zhu et al., 2019) incorporate external machine translation or summarization modules, with NCLS + MT requiring additional long-text machine translation data that is often scarce in low-resource scenarios. The ATS method (Zhu et al., 2020) depends on bilingual lexicons extracted from large parallel machine translation

Table 2
Experimental results of comparison between MCCLS and baselines in different resource scenarios.

		En2ZhSum			Zh2EnSum			Vi2ZhSum		
		R-1	R-2	R-L	R-1	R-2	R-L	R-1	R-2	R-L
	GETran	28.24	11.27	25.49	24.31	8.77	19.84	16.73	6.64	14.17
	GLTran	31.96	13.69	29.27	32.47	15.24	28.83	19.13	7.95	16.13
Minimum Scenario	**TNCLS**	21.93	6.99	18.73	19.54	5.42	16.69	16.02	5.62	13.75
	CLS + MS	22.44	7.57	19.19	20.25	6.01	17.16	16.54	6.04	14.24
	MCCLS	25.40	9.54	21.90	20.61	6.54	17.64	17.45	7.14	14.94
Medium Scenario	**TNCLS**	31.95	15.73	27.86	24.54	9.17	21.26	18.40	6.98	15.88
	CLS + MS	32.18	15.97	28.08	24.82	9.35	21.41	18.72	6.82	15.19
	MCCLS	32.77	16.46	28.49	25.25	9.86	21.95	19.21	7.07	16.21
Maximum Scenario	**TNCLS**	36.67	19.92	32.25	36.79	18.85	33.14	23.10	10.23	20.30
	CLS + MS	36.91	20.16	32.34	36.88	18.97	34.80	23.31	10.49	20.48
	MCCLS	**37.54**	**20.71**	**33.11**	**37.35**	**19.36**	**33.70**	**23.89**	**11.07**	**20.96**

datasets (e.g., LDC corpora with 208M sentence pairs), which are inaccessible for low-resource language pairs like Vietnamese-Chinese. Models like MCLAS (Bai et al., 2021) and VHM (Liang et al., 2022) necessitate auxiliary resources such as source-language summaries, increasing data dependencies beyond what is feasible under our low-resource focus. Similarly, multilingual pretrained models like mT5 (Xue et al., 2021b), while effective in resource-rich settings, are excluded because their performance gains stem from massive external corpora, introducing confounding factors that obscure the evaluation of a model's ability to learn from limited in-domain data. In contrast, we include pipeline methods such as GETran and GLTran as practical baselines, as they reflect real-world practices where practitioners resort to off-the-shelf tools (e.g., Google Translate) due to data scarcity. This approach ensures that our comparisons remain grounded in realistic low-resource constraints while highlighting the advantages of end-to-end methods like MCCLS. By maintaining this exclusion-inclusion balance, we uphold the fairness and relevance of our evaluation, directly supporting the core objective of verifying MCCLS's performance in genuine low-resource and noisy environments. Therefore, the aforementioned advanced models are not used as baselines.

5.4. Experimental results and analysis

5.4.1. Comparison between MCCLS and baselines

The experimental results are presented in Table 2. We observed that MCCLS outperformed other baseline models across all three datasets, three scenarios, and three metrics. Particularly in the Minimum Scenario, MCCLS achieved significant improvements on the En2ZhSum dataset, with ROUGE-1, ROUGE-2, and ROUGE-L increasing by 3.47, 2.55, and 3.17 compared to TNCLS. This may be due to TNCLS not being adequately trained under extremely low-resource conditions, leading to a significant drop in performance. In contrast, MCCLS maintains robust performance even with very limited training resources, demonstrating its effectiveness under extreme resource constraints.

In the Maximum Scenario, the improvements of MCCLS on the En2ZhSum and Zh2EnSum datasets were less pronounced than on the Vi2ZhSum dataset. We believe this may be because the En2ZhSum and Zh2EnSum datasets are relatively larger in scale and have higher quality levels, while the Vi2ZhSum dataset contains only 129,393 training instances and suffers from lower machine translation quality in Vietnamese, resulting in lower instance quality. MCCLS's ability to effectively leverage valuable information from low-quality instances likely contributes to its superior performance on Vi2ZhSum.

In summary, the experimental results on the En2ZhSum, Zh2EnSum, and Vi2ZhSum datasets demonstrate that our proposed MCCLS model can enhance CLS performance under resource-constrained conditions and in low-quality scenarios while maintaining performance stability.

Table 3
Ablation experiments results of MCCLS in the minimum scenario.

Model	En2ZhSum			Zh2EnSum			Vi2ZhSum		
	R-1	R-2	R-L	R-1	R-2	R-L	R-1	R-2	R-L
TNCLS	21.93	6.99	18.73	19.54	5.42	16.69	16.02	5.62	13.75
TNCLS + MD	24.78	8.99	21.29	19.96	5.87	17.03	17.07	6.79	14.57
TNCLS + DS	24.89	9.08	21.41	20.09	6.11	17.26	16.92	6.57	14.33
MCCLS	**25.40**	**9.54**	**21.90**	**20.61**	**6.54**	**17.64**	**17.45**	**7.14**	**14.94**

5.4.2. Ablation experiments

To validate the impact of our proposed multi-dimensional instance weighting and dynamic supervised signal selection on model performance, we introduced multi-dimensional instance weighting (TNCLS + MD) and dynamic supervised signal selection (TNCLS + DS) based on the baseline model TNCLS to assess the effectiveness of each module. We trained the model under the Minimum Scenario across three datasets (En2ZhSum, Zh2EnSum, and Vi2ZhSum) and compared their ROUGE scores. The experimental results are presented in Table 3, where it can be observed that TNCLS + MD, TNCLS + DS, and MCCLS all outperform the baseline model TNCLS. This indicates that the improvements proposed in this paper are effective.

5.4.3. The impact of different low-quality instance ratios

To investigate the impact of training datasets with different ratios of low-quality instances on our proposed MCCLS model and the baseline TNCLS model, we constructed datasets based on the original Zh2EnSum and Vi2ZhSum datasets. By employing a round-trip translation strategy and setting thresholds T_1 and T_2, we controlled the ratio of low-quality instances. Specifically, for Zh2EnSum, T_1 was set to 0.7 and T_2 to 0.35, while for Vi2ZhSum, T_1 was set to 0.5 and T_2 was 0.2. An instance was considered "high-quality" if the ROUGE-1 score between the back-translated summary and the original summary exceeded T_1, and the ROUGE-2 score exceeded T_2; otherwise, it was classified as "low-quality." Using this method, we constructed training datasets with low-quality instance ratios of 0 %, 10 %, 30 %, 50 %, and 70 %, each consisting of 50,000 instances.

The experimental results are presented in Tables 4. We can find that: (1) As the low-quality instance ratio increases, the performance of all models declines, but the rate of decline for MCCLS is slower than that of TNCLS. (2) For the baseline TNCLS model, high low-quality instances of 70 % led to significant performance degradation, with decreases of 4.02, 2.14, and 3.34 in ROUGE-1, ROUGE-2, and ROUGE-L scores, respectively. (3) In contrast, the MCCLS model exhibited more stable performance across datasets with varying low-quality instances ratios, showing a clear advantage in low-quality scenarios. Specifically, under a 70 % low-quality instances ratio, MCCLS achieved a 1.32

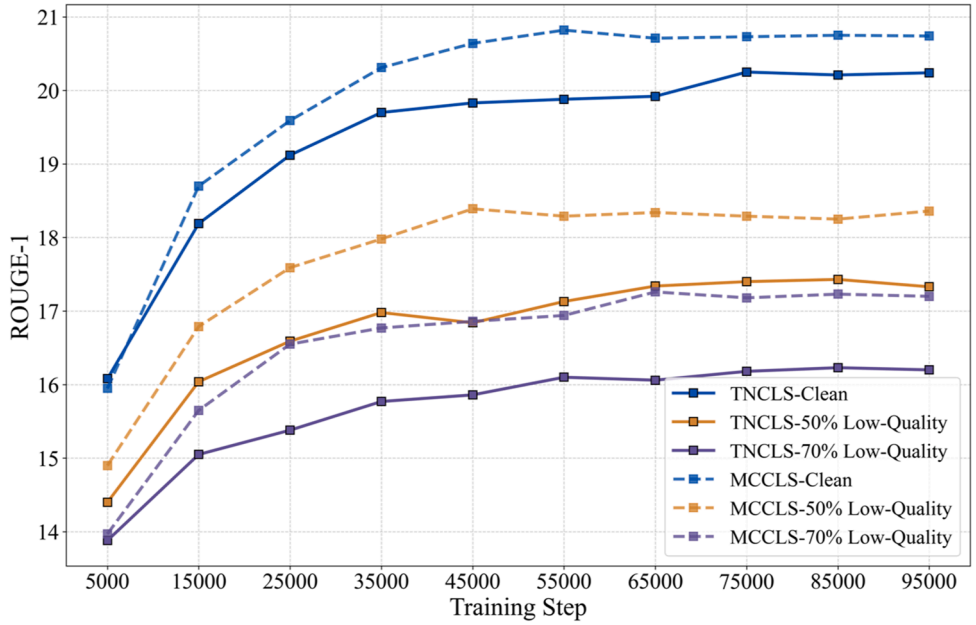


Fig. 4. The training convergence speed of TNCLS and MCCLS in different low-quality instance radios.

Table 4
Impact of different low-quality instance ratios in the minimum scenario for MCCLS on the Zh2EnSum dataset.

Low-Quality Instance Ratios	Model	Zh2EnSum		
		R-1	R-2	R-L
0%	**TNCLS**	20.25	5.73	17.12
	MCCLS	**20.82**	**6.54**	**17.81**
10%	**TNCLS**	19.78	5.51	16.83
	MCCLS	**20.39**	**6.25**	**17.40**
30%	**TNCLS**	19.18	5.32	16.14
	MCCLS	**19.87**	**5.94**	**16.85**
50%	**TNCLS**	17.43	4.35	14.70
	MCCLS	**18.39**	**5.43**	**15.63**
70%	**TNCLS**	16.23	3.59	13.78
	MCCLS	**17.26**	**5.27**	**14.85**

Table 5
Human evaluation results of MCCLS and baseline models in the minimal scenario.

Model	Zh2EnSum			Vi2ZhSum		
	IF	CC	FL	IF	CC	FL
GETran	2.75	2.62	2.79	2.66	2.58	2.82
GLTran	2.79	2.85	2.94	2.72	2.62	2.83
TNCLS	3.03	2.86	**3.35**	2.83	2.69	3.23
MCCLS	**3.09**	**3.02**	3.31	**2.86**	**2.75**	**3.25**

Table 6
GPU memory usage and training time comparison.

Model	GPU Memory Usage (M)	Training Time (s)
TNCLS	1726.83	30.62
CLS + MS	2305.56	79.91
MCCLS	1727.21	66.67

increase in the ROUGE-1 score than TNCLS. These results indicate that MCCLS is robustness to low-quality instances.

5.4.4. The convergence speed of the MCCLS model

To verify the advantages of the proposed model in terms of training convergence speed, we conducted experiments on the Zh2EnSum dataset with varying ratios of low-quality instances. As shown in Fig. 4, MCCLS not only improved performance but also achieved faster convergence compared to the baseline TNCLS model. Therefore, we conclude that MCCLS can enhance CLS performance in datasets with low-quality instances and significantly improve training efficiency.

5.5. Human evaluation

In addition to automatic evaluation, we randomly selected 50 instances from the Zh2EnSum and Vi2ZhSum test sets for human evaluation. Specifically, we recruited three graduate students proficient in both Chinese and English to evaluate the generated summaries based on three aspects: Informativeness (IF), Fluency (FL), and Conciseness (CC). Each aspect was scored on a scale of 1 (worst) to 5 (best).

The results are shown in Table 5. It can be observed that MCCLS achieves higher scores than TNCLS in both IF and FL, further demonstrating the effectiveness of our proposed method. In the Vi2ZhSum dataset, compared to Zh2EnSum, the summaries generated by models such as TNCLS and MCCLS received lower scores in IF, CC, and FL. This finding highlights the negative impact of data resource and quality limitations on the quality of model-generated summaries.

5.6. Hyperparameter sensitivity experiments

We conducted sensitivity experiments on two key hyperparameters: the confidence differentiation hyperparameter k and the dynamically supervised signal selection hyperparameter ϵ. The experimental results are shown in Fig. 5. Specifically, ROUGE scores demonstrate a slight upward trend as k increases, but stabilize when k exceeds 2. In contrast, variations in ϵ show no significant impact on ROUGE scores.

5.7. Computational effectiveness of token-level confidence estimation

To evaluate the computational effectiveness of token-level confidence estimation in MCCLS, we conducted quantitative comparisons with TNCLS and CLS + MS models under identical hardware conditions. Specifically, we compared GPU memory usage and training time across

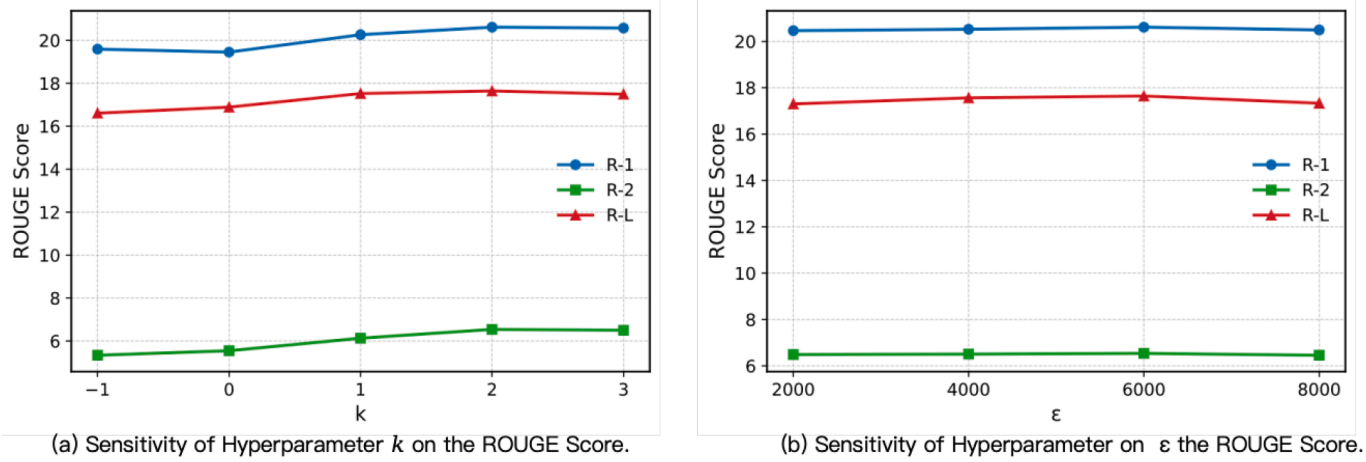


Fig. 5. Sensitivity of hyperparameter k and ε on ROUGE scores.

Example 1	Example 2
Input: time is running out for **18 miners trapped underground** after a coal mine collapsed in southern turkey on tuesday. governor murat koca said the incident occurred at the has sekerler mine near the town of ermenek in karaman province, 300 miles south of ankara close to turkey's mediterranean coast. a rescue team has been working all night and into wednesday to free the trapped miners, who are believed to be located about **350 meters below ground**. aid locals watch as rescue workers try to save 18 miners trapped after a collapse in a mine in the ermenek district in karaman, turkey. **turkish officials** say the workers' chances of survival are slim unless they managed to reach a safety gallery ...	**Input:** 中国国家发改委在周二宣布年内首次调降成品油价格。同时，并出台 成品油价格机制完善方案，包括缩短调价周期、取消 4 %调价幅度限制等 。 这标志着中国成品油市场化又向前迈进一步。 (Translation: On Tuesday, **China's** National Development and Reform Commission announced the first reduction in **refined oil prices for the year**. At the same time, it introduced a refined oil pricing mechanism improvement plan, including shortening the pricing adjustment cycle and **abolishing the 4% pricing adjustment limit**. This marks another step forward in the marketization of refined oil in China.)
Gold Summary: 土耳其一座煤矿坍塌，18 名工人被困在地下 350 米。(A coal mine in Turkey collapsed, trapping 18 workers underground at a depth of 350 meters.)	**Gold Summary:** china's domestic refined oil prices were lowered for the first time this year, and a comprehensive price formation mechanism was introduced at the same time.
TNCLS: 土耳其中部城镇班尼斯的一座桥倒塌，引发了一场大规模漏油事件，导致数百人被困。(A **bridge** in the town of Banis in central Turkey collapsed, causing a major oil spill and trapping **hundreds of people**.)	**TNCLS:** for the first time in the year , **the price adjustment of refined oil** will be **lowered by 70 - 100 %** in the year.
MCCLS: **土耳其当局称**，**矿工**被困在土耳其东南部的一个矿井里。(**Turkish authorities** said the **miners** were trapped in a mine in southeastern Turkey.)	**MCCLS:** **china lowered the price of refined oil** for the first time in the year , and **remove the 4% price adjustment range limit**.

Fig. 6. Case Study.

different models using the Zh2EnSum dataset under the minimum resource scenario. GPU memory usage refers to the average memory consumption during 1000 training steps with the same batch size 2048, while training time denotes the duration required to complete 1000 training steps with the identical batch size 2048. As shown in Table 6, MCCLS maintains a comparable GPU memory usage (1727.21M) to TNCLS (1726.83M), with a negligible increase of 0.38M parameters attributable to the lightweight design of Monte Carlo sampling-based confidence estimation. In terms of training time, MCCLS requires 66.67 seconds, which is longer than TNCLS (36.05 seconds) but shorter than CLS + MS (79.91 seconds). This moderate overhead represents a reasonable trade-off for the advantages of token-level confidence estimation,

and the efficiency improvement over CLS + MS demonstrates the practical value of our method in balancing performance and computational cost.

5.8. Case study

To provide a more intuitive demonstration of our model's performance in generating cross-lingual summaries, we conduct a case study using both the Zh2EnSum and En2ZhSum test sets. As shown in Example 1 of Fig. 6, in the En2ZhSum case, compared to the manually labeled gold summary, TNCLS generated factually incorrect content, including claims such as "hundereds of people" and "brige". It also failed to include key entity terms such as "miners" thus reducing the summary's informativeness. In contrast, the summary produced by MCCLS captures most essential information. As shown in Example 2 of Fig. 6, in the Zh2EnSum case, compared to the manually labeled gold summary, TNCLS produced incorrect information such as "lowered by 70-100 %". Moreover, important entity terms like "china" were omitted, compromising the summary's completeness. In contrast, the summary generated by MCCLS covers all key points and incorporates more critical information, such as "remove the 4 % price adjustment range limit." Overall, our method yields more accurate and information-rich English and Chinese summaries than the baseline.

6. Conclusion

This study proposes MCCLS, an innovative framework for low-resource cross-lingual summarization that effectively utilizes large-scale low-quality datasets through multi-dimensional instance weighting and dynamically supervised signal selection, achieving superior performance compared to baseline models while demonstrating remarkable stability across varying data quality ratios and faster convergence, as validated through extensive experiments on En2ZhSum, Zh2EnSum and Vi2ZhSum datasets, human evaluations, and case studies showing more accurate summary generation.

CRediT authorship contribution statement

Yongbing Zhang: Writing – original draft, Methodology, Investigation, Conceptualization; **Fang Nan:** Writing – review & editing, Validation, Methodology; **Shengxiang Gao:** Writing – review & editing, Validation; **Yuxin Huang:** Writing – review & editing, Visualization, Validation; **Kaiwen Tan:** Writing – review & editing, Supervision, Resources, Formal analysis, Conceptualization; **Zhengtao Yu:** Writing – review & editing, Supervision, Validation, Formal analysis.

Data availability

Data will be made available on request.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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